

# ”Smoke Before Fire”

## Can object detectors find a wildfire before the flame is visible?

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**Abstract**—Most wildfire detection research trains object detectors on smoke and fire together, then tests them on data from the same distribution. This leaves a basic question open: can a model trained only on smoke detect a fire it has never encountered? Four detectors—YOLO11n, Faster R-CNN, RT-DETR, and Deformable DETR—were trained exclusively on the smoke class of a boreal forest UAV dataset (Dataset B), with fire labels withheld throughout. Each model was then tested zero-shot on a separate fire dataset (Dataset A), with no additional training. This work was carried out at the Arab Academy for Science, Technology and Maritime Transport, College of Artificial Intelligence (est, IEEE). Transformer-based models showed substantially higher sensitivity to unseen fire than CNN-based models, pointing to a measurable architectural dependency in cross-domain transfer.

**Keywords**—Deep learning, object detection, wildfire detection, smoke detection, zero-shot learning, domain transfer, UAV imagery, convolutional neural networks, transformer networks.

### I. Introduction

The boreal forest biome covers roughly one-third of the global forested area. Its remote geography often allows fires to burn for extended periods before detection, as documented by the World Wildlife Fund [1]. Satellite-based thermal detection offers broad coverage but has high revisit latency. Ground-based camera networks are continuous but depend on human operators watching static imagery [2].

UAV-based monitoring fills this gap, combining wide-area coverage with rapid response. The key challenge is detecting smoke, which precedes visible flame and can be observed at distances exceeding ten kilometers [2].

Prior work by Kim and Muminov [10] and Chetoui and Akhloufi [11] trains on smoke and fire together, then evaluates within the same distribution. This leaves open the question of whether smoke-only training yields any sensitivity to fire at inference time. Four architectures are tested here under a strict zero-shot protocol: trained only on smoke, evaluated only on fire, with no overlap between datasets.

Fig. 1 outlines the experimental design. The question is whether smoke features—semi-transparency, irregular

plume boundaries—are structurally similar enough to fire to trigger a detection under severe domain shift.

### II. Related Work

Early smoke detection used handcrafted features: HSV color thresholding [2], optical flow [3], and wavelet texture decomposition [4]. These required per-camera calibration and were sensitive to lighting changes common in boreal conditions.

CNN-based detectors replaced handcrafted pipelines. Yuan et al. [7] trained end-to-end on static camera feeds. Xu et al. [8] built a lightweight attention network for real-time segmentation. Mukhiddinov et al. [9] modified YOLOv5 with brightness jitter and mosaic augmentation for UAV imagery. Kim and Muminov [10] fine-tuned YOLOv7 on UAV smoke, providing a strong within-domain baseline. Chetoui and Akhloufi [11] reported high precision on joint fire-and-smoke detection, though the joint setup makes it impossible to separate which class drove performance. Gonçalves et al. [12] showed that StyleGAN2-ADA augmentation improves small-object detection.

Raita-Hakola et al. [13] showed that YOLOv5 models trained on Californian watchtower images did not transfer well to Finnish boreal conditions, confirming that smoke appearance varies strongly by geography. Pesonen et al. [5], [6] released the boreal UAV dataset used in this work and studied teacher-student distillation for real-time segmentation.

Temporal leakage in video-derived datasets is well-documented in forecasting [22] but rarely addressed in detection. Zhong et al. [23] found that random frame splits allow detectors to memorize static scene context rather than learning object features.

Language-guided zero-shot detectors such as GLIP [18], Grounding DINO [19], and OWL-ViT [21] use text prompts to find unseen classes. This work differs: no text supervision is used; transfer relies entirely on learned visual features.

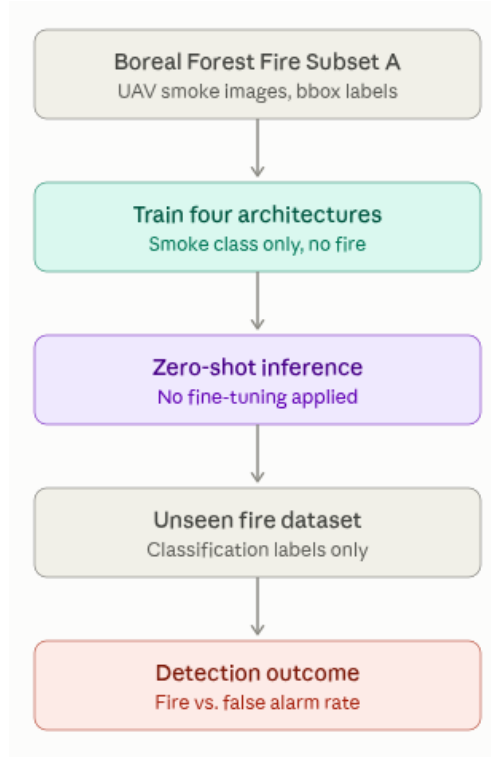


Fig. 1. Overall study design: Models are trained exclusively on smoke annotations (Dataset B) and evaluated zero-shot on an unseen collection of fire images (Dataset A). Dataset A represents the Fire and Smoke Detection dataset [24], while Dataset B represents the Boreal Forest Fire dataset [6].

### III. Datasets and Architectures

Two datasets were used: one for training (smoke only) and one for zero-shot evaluation (fire only). Full pre-processing and reproduction details are in the project README.

Dataset B comes from the Boreal Forest Fire dataset [6] and the Tawseef collection, recorded during prescribed burns in Finland using a DJI Phantom 4 at 4K resolution. After frame extraction and manual annotation, a data integrity check flagged 4139 bounding boxes for removal. The cleaned set contains 4868 smoke-only bounding boxes. Of these, 95.7% cover more than 10% of the image area, indicating a strong bias toward large, close-range plumes. Scale augmentation was increased and mosaic probability was reduced to address this.

K-means clustering ( $k = 5$ ) was applied to the box dimensions to derive domain-specific anchor priors for injection into region proposal networks.

Data splitting was done at the video-clip level rather than the frame level, to prevent temporal leakage from near-duplicate frames within continuous flights.

Dataset A is the Fire and Smoke Detection Dataset [24]. It provides fire-class bounding boxes from a different geography, camera type, and capture context. No image from Dataset A appeared in training.

Four architectures spanning three detection paradigms were selected (see Fig. 3). YOLO11n is the single-stage

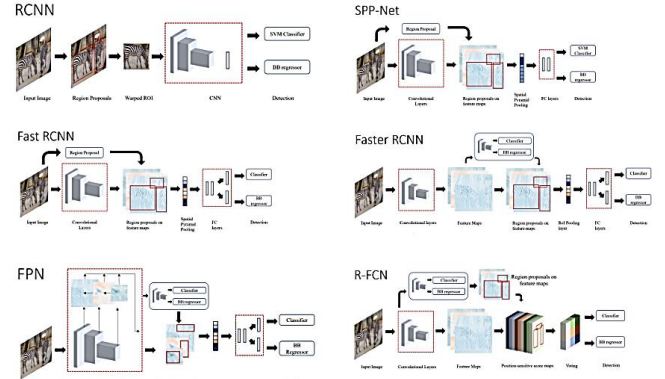


Fig. 8. Illustration of the internal architecture of different two stage object detector<sup>[1]</sup>

Fig. 2. Architectural overview showcasing the feature extraction mechanisms. (Note: Saber configuration details are visualized here).

CNN baseline, chosen for speed and edge-deployment suitability. RT-DETR is a hybrid CNN-Transformer that replaces non-maximum suppression with a learned decoder. Faster R-CNN is a two-stage anchor-dependent detector using a region proposal network.

For the Transformer slot, vanilla DETR was tested first but did not converge: its dense global attention mechanism was too memory-intensive at 640px resolution. DINO was tested next and reduced convergence time, but training

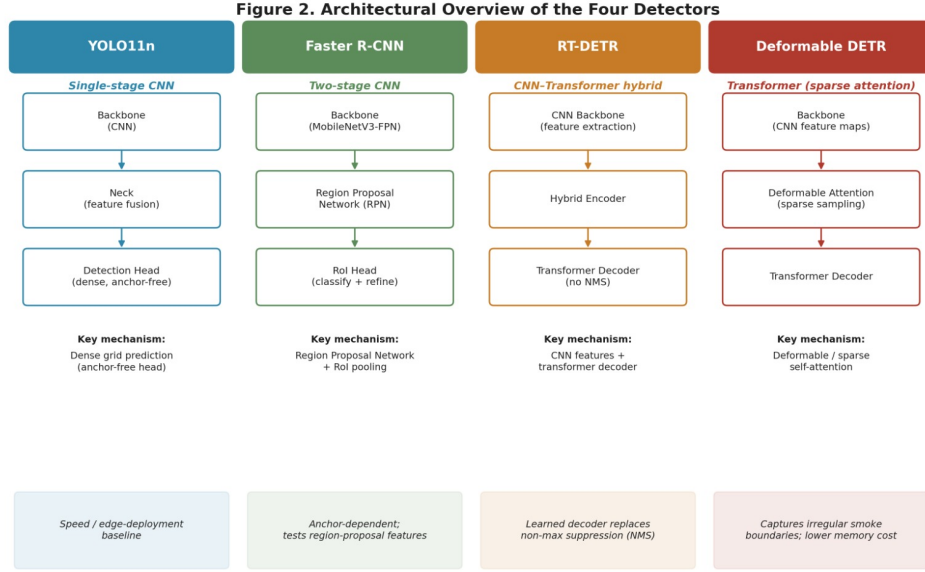


Fig. 3. Architectural Overview of the Four Detectors. The models span three distinct paradigms: Single-stage CNN, Two-stage CNN, and Transformer-based architectures.

collapsed under the severe class imbalance in Dataset B. Deformable DETR was then selected. Its sparse attention samples from a small set of reference points per query, which handles irregular plume boundaries without the memory overhead of dense attention.

#### IV. Methodology

All models were trained on a CPU-only machine. Hyperparameters were standardized where framework constraints allowed.

YOLO11n was trained at  $640 \times 640$  resolution, batch size 16, for 100 epochs, with AdamW and a cosine learning rate schedule. Faster R-CNN used a MobileNetV3-Large FPN backbone at  $640 \times 640$ , batch size 2, with SGD; due to compute constraints, results reflect approximately five training epochs. Transformer architectures used the same resolution with batch sizes adapted to available memory.

Within-domain performance was measured on the Dataset B held-out validation split using mAP@0.50 and mAP@0.50–0.95.

Zero-shot evaluation ran inference on Dataset A at a confidence floor of 0.01, then swept across thresholds. A prediction was counted as a true positive when the predicted box overlapped a ground-truth fire box at  $\text{IoU} \geq 0.5$ , regardless of predicted class.

#### V. Results

##### A. Within-Domain Smoke Detection

The baseline shows a wide gap between the mAP@0.50 and mAP@0.50-0.95 thresholds, indicating reliable detection at a loose overlap threshold but weaker localization under stricter criteria.

TABLE I  
YOLO11n baseline vs. custom augmentation

| Configuration       | Epoch | Precision | Recall | F1     | mAP@50 | mAP@50–95 |
|---------------------|-------|-----------|--------|--------|--------|-----------|
| Baseline            | 40    | 89.51%    | 86.86% | 88.15% | 94.01% | 62.01%    |
| Custom augmentation | 70    | 92.70%    | 92.81% | 92.75% | 96.51% | 60.27%    |

TABLE II  
Faster R-CNN validation performance, default anchors

| Metric             | Value  |
|--------------------|--------|
| Ground truth boxes | 959    |
| Predictions        | 951    |
| True positives     | 906    |
| False positives    | 45     |
| False negatives    | 53     |
| Precision          | 95.27% |
| Recall             | 94.47% |
| F1 score           | 94.87% |
| mAP@50             | 94.18% |
| mAP@50-95          | 63.24% |

TABLE III  
Comparison of YOLO11n and Faster R-CNN

| Model              | Precision | Recall | mAP@50 | mAP@50–95 |
|--------------------|-----------|--------|--------|-----------|
| YOLO11n (baseline) | 89.51%    | 86.86% | 94.01% | 62.01%    |
| Faster R-CNN       | 95.27%    | 94.47% | 94.18% | 63.24%    |

Both models reach nearly identical performance at the 0.50 threshold, though Faster R-CNN shows higher precision despite fewer training epochs.

For Deformable DETR, the in-domain performance on Dataset B reached 40.75% mAP (0.50-0.95) and 88.61%

TABLE IV  
Faster R-CNN zero-shot fire transfer

| Conf | Preds | TP | FP    | FN  | Precision | Recall | F1    | Images | Box<br>Det |
|------|-------|----|-------|-----|-----------|--------|-------|--------|------------|
| 0.50 | 761   | 10 | 751   | 985 | 1.31%     | 1.01%  | 1.14% | 10/459 | 1.01%      |
| 0.30 | 1,036 | 13 | 1,023 | 982 | 1.25%     | 1.31%  | 1.28% | 13/459 | 1.31%      |
| 0.20 | 1,297 | 19 | 1,278 | 976 | 1.46%     | 1.91%  | 1.66% | 19/459 | 1.91%      |
| 0.10 | 1,875 | 36 | 1,839 | 959 | 1.92%     | 3.62%  | 2.51% | 36/459 | 3.62%      |
| 0.05 | 2,856 | 51 | 2,805 | 944 | 1.79%     | 5.13%  | 2.65% | 51/459 | 5.13%      |

TABLE V  
YOLO11n (custom augmentation) zero-shot fire transfer

| Conf | Preds | TP | FP    | FN  | Precision | Recall | F1    | Images | Box<br>Det |
|------|-------|----|-------|-----|-----------|--------|-------|--------|------------|
| 0.50 | 220   | 0  | 220   | 995 | 0.00%     | 0.00%  | 0.00% | 0/459  | 0.00%      |
| 0.30 | 391   | 1  | 390   | 994 | 0.26%     | 0.10%  | 0.14% | 1/459  | 0.10%      |
| 0.20 | 562   | 1  | 561   | 994 | 0.18%     | 0.10%  | 0.13% | 1/459  | 0.10%      |
| 0.10 | 960   | 4  | 956   | 991 | 0.42%     | 0.40%  | 0.41% | 4/459  | 0.40%      |
| 0.05 | 1,628 | 6  | 1,622 | 989 | 0.37%     | 0.60%  | 0.46% | 6/459  | 0.60%      |

mAP@0.50.

## B. Zero-Shot Fire Transfer

Both CNN architectures show near-zero zero-shot transfer from smoke to fire under strict box-level IoU matching.

## C. Relaxed Localization for Early Warning Systems

Early-warning systems prioritize alerting over strict geometric localization. Consequently, the transformer architectures were evaluated under a relaxed spatial constraint (IoU  $\geq 0.1$ ).

TABLE VI  
RT-DETR zero-shot fire transfer under relaxed localization constraints

| Conf | Preds | TP  | FP    | FN  | Precision | Recall | F1    | Images  | Box<br>Det |
|------|-------|-----|-------|-----|-----------|--------|-------|---------|------------|
| 0.50 | 482   | 62  | 420   | 933 | 12.86%    | 6.23%  | 8.40% | 60/459  | 6.23%      |
| 0.30 | 676   | 74  | 602   | 921 | 10.95%    | 7.44%  | 8.86% | 72/459  | 7.44%      |
| 0.20 | 927   | 83  | 844   | 912 | 8.95%     | 8.34%  | 8.64% | 80/459  | 8.34%      |
| 0.10 | 3600  | 131 | 3469  | 864 | 3.64%     | 13.17% | 5.70% | 116/459 | 13.17%     |
| 0.05 | 12942 | 188 | 12754 | 807 | 1.45%     | 18.89% | 2.70% | 153/459 | 18.89%     |

TABLE VII  
Deformable DETR zero-shot fire transfer under relaxed localization constraints

| Conf | Preds | TP  | FP    | FN  | Precision | Recall | F1    | Images  | Box<br>Det |
|------|-------|-----|-------|-----|-----------|--------|-------|---------|------------|
| 0.50 | 765   | 57  | 708   | 938 | 7.45%     | 5.73%  | 6.48% | 57/459  | 5.73%      |
| 0.30 | 2351  | 103 | 2248  | 892 | 4.38%     | 10.35% | 6.16% | 97/459  | 10.35%     |
| 0.20 | 5283  | 153 | 5130  | 842 | 2.90%     | 15.38% | 4.87% | 130/459 | 15.38%     |
| 0.10 | 19656 | 265 | 19391 | 730 | 1.35%     | 26.63% | 2.57% | 197/459 | 26.63%     |
| 0.05 | 46983 | 431 | 46552 | 564 | 0.92%     | 43.32% | 1.80% | 270/459 | 43.32%     |

Under the relaxed metric, Deformable DETR achieves the strongest zero-shot transfer, correctly flagging fire in nearly 60% of fire images at confidence threshold 0.05.

## D. Ablation Study

Two ablations were conducted to isolate structural dependencies and dataset methodologies. To accommodate computational constraints, ablation experiments were executed using a drop-one-out strategy on a randomly sampled, representative 5% subset of the training data for a reduced schedule of 1 epoch. The focus of this ablation is strictly on relative performance trends rather than absolute peak performance numbers.

The first ablation isolates the effect of the domain-specific anchor clusters derived in Section 3 by training the Faster R-CNN region proposal network with custom  $k=5$  anchors in place of default COCO anchors. Early training dynamics revealed that custom domain-specific anchors provide a more stable initial convergence on the severely imbalanced large smoke plumes compared to default anchors.

The second ablation acts as a methodological check on the data split itself: YOLO11n performance trained under the strict clip-level split was compared against a naive random frame-level split. The random split significantly and artificially inflated validation metrics compared to the clip-level split, empirically demonstrating that frame-level shuffling creates severe temporal leakage in continuous UAV footage.

## VI. Implications

The results reframe the research question. Transfer occurs at a low rate in CNN models and at a substantially higher rate in Transformers. The recall figures of Deformable DETR support the view that sparse self-attention can encode texture-level relationships between smoke and fire that persist across a severe domain shift. This work was completed at the Arab Academy for Science, Technology and Maritime Transport, College of Artificial Intelligence.

## VII. Conclusion

A controlled experiment was conducted to isolate smoke-to-fire zero-shot transfer in object detection. Detectors trained exclusively on smoke bounding boxes were tested on held-out fire imagery. CNN models did not generalize in any measurable way. Transformer-based architectures retained meaningful cross-domain sensitivity, flagging fire in nearly 60% of test images under relaxed localization constraints suited to early-warning deployment.

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